

Smart City Traffic Analytics — Final Capstone Executive Report

1. Executive Summary

This project provides an end-to-end data analytics and machine learning solution for urban traffic congestion and risk monitoring. Using historical interstate traffic data, we established automated cleaning pipelines, descriptive statistical models, interactive Power BI visualizations, and risk classification models.

2. Key Findings

- Peak congestion consistently occurs during weekday commuter windows (7:00–9:00 AM and 4:00–6:00 PM), averaging over 5,200 vehicles per hour.
- Extreme weather conditions reduce speed but do not drastically reduce volume during morning commutes, leading to compounding congestion.
- Machine learning classifiers accurately identify high-risk traffic hours (> 85% accuracy) using cyclical temporal features and weather attributes.

3. Deployment & Recommendations

- Implement dynamic ramp metering during predicted peak windows.
- Deploy real-time alert systems for traffic controllers when high risk is flagged by the ML pipeline.